3DWEBSOFT INC. FOUNDATION

SCHOOL FUTURE TECH PROGRAMM

Faculty Name :

Course Duration	: 5 Months (150 Days)
Classes	: Offline & Online
Daily Teaching	: 3 Hours
Total Hours	: 300 Hours

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LEVEL 1 – JUNIOR INNOVATORS (5th–6th Standard)

MODULE 1 – COMPUTER BASICS (Classes 1–20)

Class 2 — History of Computers & Generations of Computers (Complete Notes)

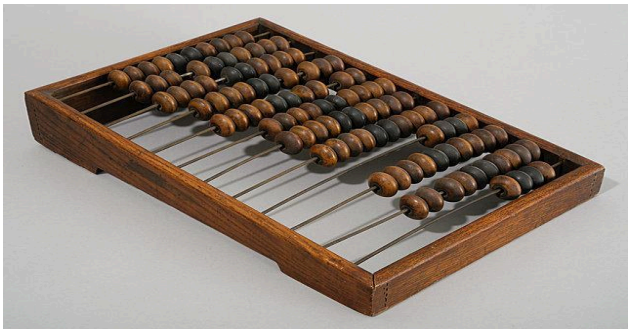
Complete Theory & Practical Notes

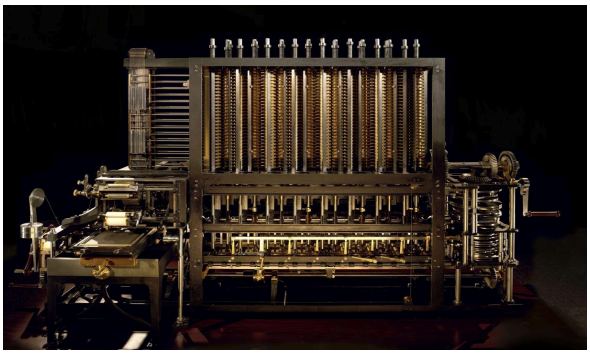
Learning Objectives

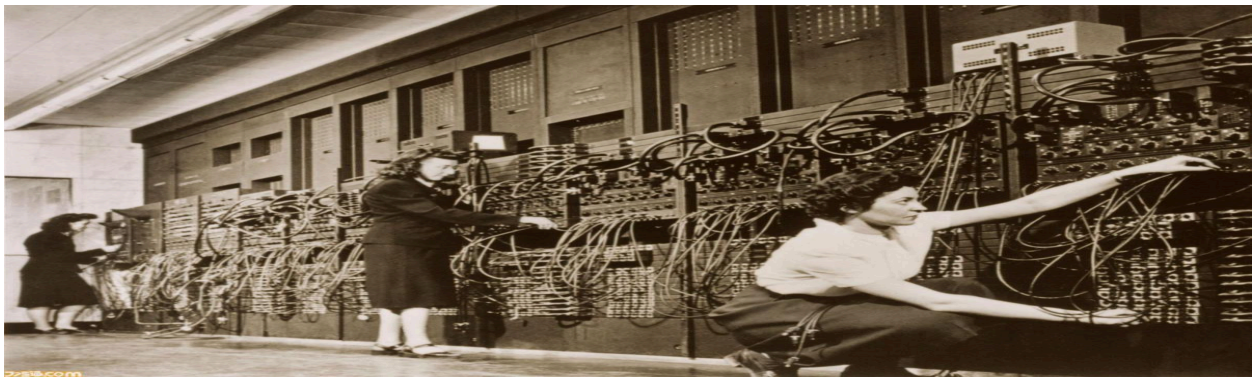
At the end of this class, students will be able to:

- Understand how computers developed over time.
- Learn the five generations of computers.
- Identify old and modern computers.
- Create a computer timeline.
- Use Scratch to build a simple interactive timeline.

History of Computers:-







What is the History of Computers?

The history of computers tells us how machines changed from simple counting tools into today's powerful smart computers.

Thousands of years ago, people counted using stones and fingers. Later, they invented machines to make calculations easier. Today, computers help us study, play games, create art, communicate, and even use Artificial Intelligence (AI).

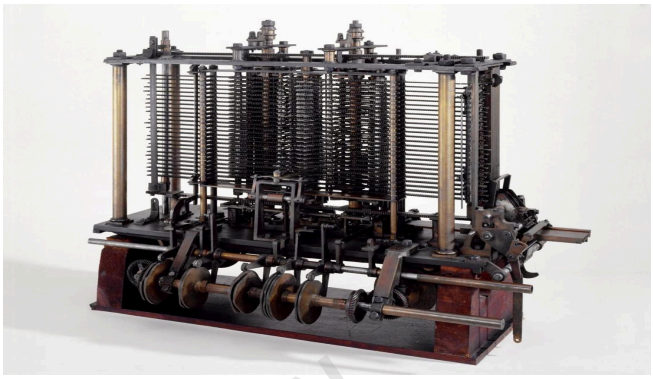
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Timeline of Computer History

Year	Invention	Inventor
3000 BC	Abacus	Ancient Chinese
1642	Pascaline	Blaise Pascal
1801	Punch Card Machine	Joseph Jacquard
1837	Analytical Engine	Charles Babbage
1890	Tabulating Machine	Herman Hollerith
1946	ENIAC	John Mauchly & J. Presper Eckert
Today	AI Computers	Modern Scientists

Father of Computer







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Charles Babbage

- Born in England.
- Designed the ***Analytical Engine.***
- Known as the ***Father of Computers.***
- His ideas inspired modern computers.

Interesting Facts

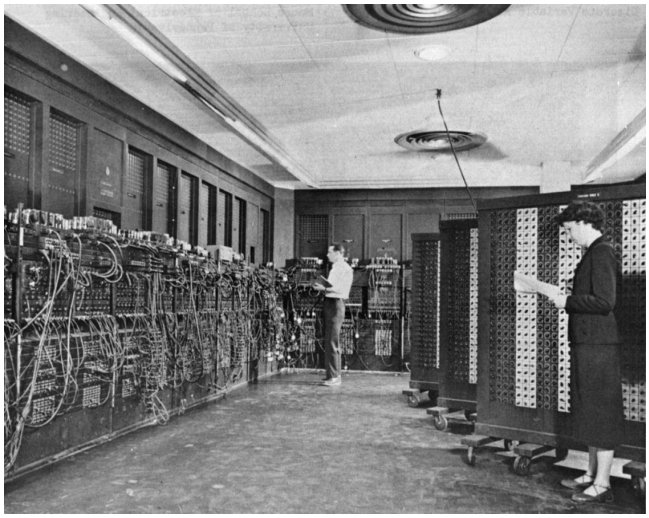
The first computers were as big as a classroom.

Early computers produced a lot of heat.

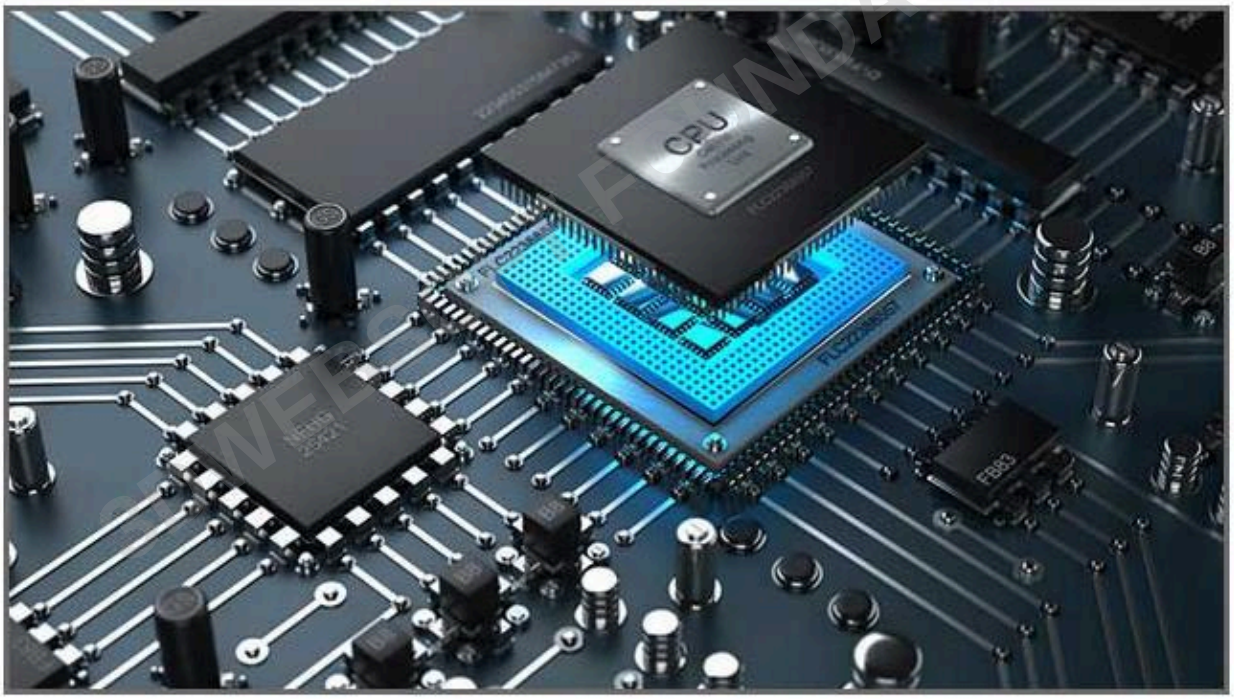
They consumed huge amounts of electricity.

Today's laptops are thousands of times faster.

Generations of Computers







Computers have improved in five generations.

First Generation (1940–1956)

Technology

Vacuum Tubes

Features

- Very large
- Very expensive
- Slow
- Produced lots of heat
- Used machine language

Example:-

- ENIAC

Second Generation (1956–1963)

Technology

Transistors

Features

- Smaller
- Faster
- Less heat
- More reliable

Example:-

IBM 1401

Third Generation (1964–1971)

Technology

Integrated Circuits (IC)

Features

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- Faster
- Smaller
- Keyboard and Monitor introduced

Example:-

IBM System/360

Fourth Generation (1971–Present)

Technology

Microprocessors

Features

- Personal Computers
- Laptop
- Mobile Phones
- Tablets

Examples:-

- Desktop
- Laptop
- Smartphone

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Fifth Generation (Present & Future)

Technology

Artificial Intelligence (AI)

Features

- Voice Assistant
- Face Recognition
- Robots
- Self-driving Cars
- Smart Chatbots

Examples:-

- ChatGPT
- Alexa
- Google Assistant

Computer Generations Overview

Computer Generations Timeline

Approximate time periods of the five computer generations.

05501,1001,6502,2001st2nd3rd4th5th

Classroom Activity:-

Timeline Chart Activity

Materials Needed

- A4 Paper
- Scale
- Pencil
- Colors

Students draw a horizontal timeline.

Example:-

Abacus
↓
Pascaline
↓
Analytical Engine
↓
ENIAC
↓
Desktop
↓
Laptop
↓
AI Computer

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Practical Coding Activity (Scratch)

Project Name

Computer History Timeline

Learning Outcome

Students create an interactive timeline where clicking each computer shows information.

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Step 1

Open Scratch.

Create a new project.

Delete the Cat sprite.

Step 2

Add Sprites

- Abacus
- Old Computer
- Desktop
- Robot

Add a simple background.

Step 3

Create Buttons

- Next
- Previous

Step 4

Programming Logic

When Green Flag Clicked

when green flag clicked

show

say "Welcome to Computer History!" for 3 seconds

When Abacus Clicked

when this sprite clicked

say "I am the first counting tool." for 3 seconds

When ENIAC Clicked

when this sprite clicked

say "I am the first electronic computer." for 3 seconds

When Desktop Clicked

when this sprite clicked

say "I belong to the Fourth Generation." for 3 seconds

When Robot Clicked

when this sprite clicked

say "I represent Artificial Intelligence." for 3 seconds

Advanced Scratch Challenge

Create a ***Quiz.***

Ask:

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Which generation uses AI?

1

2

3

4

5

If answer = **5**

say "Excellent!"

Else

say "Try Again!"

AI Connection

Ask students:

- Can robots think?
- Where have you seen AI?
- Have you used Google Assistant or ChatGPT?
- Can AI draw pictures?

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Fun Quiz

1. Who is called the Father of Computers?
    - A) Newton
    - **B) Charles Babbage**
    - C) Einstein
2. Which was the first counting machine?
    - A) Laptop
    - **B) Abacus**
    - C) Tablet
3. Which generation uses AI?
    - A) First
    - B) Third

    - **C) Fifth**
4. Which technology is used in the Fourth Generation?
    - A) Vacuum Tubes
    - **B) Microprocessor**
    - C) IC Chips

Homework

- Draw the five generations of computers.
- Write two features of each generation.
- Ask your grandparents what computers looked like when they were young.
- Build the Scratch timeline at home and add one extra computer image.

Learning Outcome

Students will be able to:

- Explain the history of computers.
- Name the five computer generations.
- Identify technologies used in each generation.
- Create a timeline chart.
- Develop a simple interactive Scratch project demonstrating computer history.

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